



**PUDUCHERRY TECHNOLOGICAL UNIVERSITY
PUDUCHERRY**

B.TECH DEGREE EXAMINATION NOVEMBER 2024

INFORMATION TECHNOLOGY

FIFTH SEMESTER

ITA116 COMPUTER NETWORKS

Time: Three Hours

Max. Marks: 75

Part A - Answer ALL Questions

5 x 2 = 10 Marks

No.	Question	CO1
i.	Define Computer Networks.	CO2
ii.	What is HTTP?	CO3
iii.	Define Multiplexing and Demultiplexing.	CO5
iv.	What is IPV6?	CO5
v.	Differentiate between Bridges and Switches.	CO1

Part B - Answer ALL Questions

5 x 10 = 50 Marks

No.	Question	Marks	CO
1.	a. Distinguish between Connection oriented and connectionless service.	(5)	CO1
	b. Explain about Circuit Switching and Packet Switching.	(5)	CO1
	(OR)		
2.	a. Discuss about Internet Protocol Stack.	(5)	CO1
	b. Write short notes on Computer Networking and the Internet.	(5)	CO1
3.	Describe the role of a Domain Name Service on a computer network with reference to its components.	(10)	CO2

(OR)

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| 4. | Explain in detail about RTCP and RTSP. | (10) | CO2 |
| 5. | a. What is UDP? Explain the UDP header format and its applications in detail. | (5) | CO3 |
| | b. Explain the concept of Go-Back-N and selective repeat. | (5) | CO3 |
| | (OR) | | |
| 6. | a. Briefly explain about TCP and list out the advantages and disadvantages. | (5) | CO3 |
| | b. Discuss about flow control and congestion control. | (5) | CO3 |
| 7. | Explain the function of ARP & DHCP. | (10) | CO5 |
| | (OR) | | |
| 8. | a. Explain OSPF routing protocol and its router types. | (5) | CO5 |
| | b. How to avoid congestion in network layer? Explain. | (5) | CO5 |
| 9. | Explain the concept of Link Level Flow Control. | (10) | CO1 |
| | (OR) | | |
| 10. | With a neat diagram explain about Token Ring and FDDI. | (10) | CO1 |

Part C - Answer ALL Questions

1 x 15 = 15 Marks

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| 11. | With a neat sketch, explain the architecture of OSI Reference model. | (15) | CO1 |
| | (OR) | | |
| 12. | Briefly explain about SSL and TLS. Compare it. | (15) | CO4 |



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**B.TECH DEGREE EXAMINATION NOVEMBER 2024
INFORMATION TECHNOLOGY**

**FIFTH SEMESTER
ITA203 INTRODUCTION TO BUSINESS INTELLIGENCE**

Time: Three Hours

Max. Marks: 75

Part A - Answer ALL Questions

5 x 2 = 10 Marks

No.	Question	CO
i.	What is Business Intelligence? Give its benefits.	CO1
ii.	Give any two real time examples for violation of digital ethics.	CO2
iii.	What is meant by Outlier Analysis?	CO3
iv.	Define: B2B model. Give an example.	CO1
v.	List out various applications of Machine learning algorithms.	CO3

Part B - Answer ALL Questions

5 x 10 = 50 Marks

No.	Question	Marks	CO
1.	Briefly Describe about Business Intelligence Architecture with neat diagram. Give different types of BI tools and software. (OR)	(10)	CO1
2.	Explain about various types of Business Intelligence users and give their roles and responsibilities.	(10)	CO1
3.	Discuss how Data Discovery process works in BI? Give various benefits of Data Discovery. (OR)	(10)	CO2
4.	Explain about Artificial Intelligence techniques and its different stages of AI developments. Discuss various AI training models in detail.	(10)	CO2

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| 5. | Explain about CCR model in detail. | (10) | CO3 |
| (OR) | | | |
| 6. | Briefly discuss about Cluster Analysis in detail. Explain various applications of Cluster Analysis. | (10) | CO3 |
| 7. | Explain in detail about marketing models in Business Intelligence Applications. | (10) | CO1 |
| (OR) | | | |
| 8. | What is meant by Production model? Explain in detail. | (10) | CO1 |
| 9. | Explain about various emerging technologies of BI in detail. | (10) | CO3 |
| (OR) | | | |
| 10. | Briefly discuss about machine learning techniques in Business Intelligence Process. | (10) | CO1 |

Part C - Answer ALL Questions

1 x 15 = 15 Marks

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| 11. | a. Discuss the future of Business Intelligence with real time applications. | (8) | CO3 |
| | b. Explain in detail about BI search. | (7) | CO3 |
| (OR) | | | |
| 12. | Describe different Predictive Analytics models and techniques in detail. | (15) | CO2 |



**PUDUCHERRY TECHNOLOGICAL UNIVERSITY
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B.TECH DEGREE EXAMINATION DEC 2024

ACE/BME/PCE/IT/EEE/ECE

FOURTH/FIFTH SEMESTER

CSA301 INTRODUCTION TO PYTHON PROGRAMMING

Time: Three Hours

Max. Marks: 75

Part A - Answer ALL Questions

5 x 2 = 10 Marks

No.	Question	CO
i.	List out python keywords.	CO1
ii.	Give an example for identity operators in python.	CO2
iii.	What are the difference between tuple and list object.	CO3
iv.	What is lambda function? Give an example.	CO4
v.	What is the purpose of exception handling?	CO5

Part B - Answer ALL Questions

5 x 10 = 50 Marks

No.	Question	Marks	CO
1.	a. Enumerate the important features of python.	(5)	CO1
	b. Write steps to install python in a computer.	(5)	CO1
	(OR)		
2.	a. What are the features of interactive mode of programming?	(5)	CO1
	b. Write a python script to get 3 numbers and print them in reverse order of feeding.	(5)	CO1
3.	a. Describe in detail about logical and bitwise operators.	(5)	CO2
	b. What are the places to use membership operator? Give examples.	(5)	CO2
	(OR)		
4.	a. Describe the variants of for and while loops in python.	(5)	CO2
	b. Write a python script to sort in descending N numbers without using built-in function.	(5)	CO2

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| 5. | a. | Describe the various built-in Tuple's functions & Methods. | (5) | CO3 |
| | b. | Write a python script to create and access a dictionary. | (5) | CO3 |
| (OR) | | | | |
| 6. | a. | Write a program to describe create, access , update and delete of a List. | (5) | CO3 |
| | b. | Write a python script to access modules and functions of a package. | (5) | CO3 |
| 7. | a. | Explain how to define a function and call that function in python | (5) | CO4 |
| | b. | Describe the various Scope of the Variables in a Function with examples. | (5) | CO4 |
| (OR) | | | | |
| 8. | a. | Describe about name spacing in python. | (5) | CO4 |
| | b. | Explain how to install PIP in an installed python. | (5) | CO4 |
| 9. | a. | Write a script to define a student class and create 5 students objects. | (5) | CO5 |
| | b. | Describe how to handle the constructor and destructors in python. | (5) | CO5 |
| (OR) | | | | |
| 10. | a. | Write a python scripts to store a paragraph on a text file. | (5) | CO5 |
| | b. | Describe the various built in function of file handling in python. | (5) | CO5 |

Part C - Answer ALL Questions

1 x 15 = 15 Marks

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| 11. | a. | Write a python script to implement multiplication and addition of two matrices. | (8) | CO3 |
| | b. | Write a python script to create a namelist in sorted order without using build-in function. | (7) | CO4 |
| (OR) | | | | |
| 12. | a. | Write necessary python scripts to create a package. | (7) | CO3 |
| | b. | Write a python script to find greatest and lowest of n numbers in an array and exchange these number's position. | (8) | CO4 |



**PUDUCHERRY TECHNOLOGICAL UNIVERSITY
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**B.TECH DEGREE EXAMINATION NOVEMBER 2024
INFORMATION TECHNOLOGY**

**FIFTH SEMESTER
ITA117 INFORMATION CODING TECHNIQUES**

Time: Three Hours

Max. Marks: 75

Part A - Answer ALL Questions

5 x 2 = 10 Marks

No.	Question	CO
i.	Write the equation for self-information with respect to probability.	CO1
ii.	Define Run-length encoding.	CO3
iii.	What is meant by quantization?	CO4
iv.	List of various types of frames used in video compression and write their characteristics.	CO5
v.	Define hamming distance. Give example.	CO2

Part B - Answer ALL Questions

5 x 10 = 50 Marks

No.	Question	Marks	CO
1.	a. In a binary PCM if '0' occur with probability 1/4 and '1' occur with probability 3/4, then calculate the amount of information conveyed by each binit. b. State the properties of Entropy.	(5) (5)	CO1 CO1
	(OR)		
2.	Explain the concepts of Coding theory, code efficiency and redundancy.	(10)	CO1
3.	Explain Shannon-fano coding algorithm with an example problem.	(10)	CO4
	(OR)		
4.	Using Lempel-Ziv Algorithm, code the String 1010110110101011 and develop a Dictionary.	(10)	CO4

5. Explain the JPEG image compression standard with neat architecture. ¹²
(5) CO5
- (OR)
6. Describe the concept of Linear Predictive coding with diagram. (10) CO4
7. Explain the concept of motion estimation and compensation used in video compression. (10) CO3
- (OR)
8. Describe H.261 Encoding procedure with suitable diagram. (10) CO4
9. Explain Linear Block codes with an example. (10) CO2
- (OR)
10. What are BCH codes? Write the procedure for the construction of BCH codes. (10) CO5

Part C - Answer ALL Questions

1 x 15 = 15 Marks

11. With the following symbols and their probability of occurrence, encode the message "went#" using arithmetic coding.

Symbols	E	N	T	W	#
Probability	0.3	0.3	0.2	0.1	0.1

(15) CO5

(OR)

12. Give a comparative study on MPEG-1, MPEG-2 and MPEG-4. (15) CO4



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**B.TECH DEGREE EXAMINATION NOVEMBER 2024
INFORMATION TECHNOLOGY**

**FIFTH SEMESTER
ITA114 DATABASE MANAGEMENT SYSTEMS**

Time: Three Hours

Max. Marks: 75

Part A - Answer ALL Questions

5 x 2 = 10 Marks

No.	Question	CO
i.	List the limitations/disadvantages of DBMS.	CO1
ii.	What is Data Model?	CO2
iii.	What are the characteristics of SQL?	CO5
iv.	What is Normalization? What are the types of Normal Forms?	CO4
v.	List out the states of a Transaction.	CO3

Part B - Answer ALL Questions

5 x 10 = 50 Marks

No.	Question	Marks	CO
1.	a. What are the problems with File system data management?	(4)	CO1
	b. Explain in detail the components of DB system environment. (OR)	(6)	CO1
2.	a. Differentiate between DBMS and RDBMS	(4)	CO1
	b. With suitable example explain Entities and Attribute.	(6)	CO1
3.	a. Differentiate Network model with Hierarchical model.	(5)	CO2
	b. Describe the Relational Model in detail with an example.	(5)	CO2

(OR)

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| 4. | a. | Write a note on Database Languages. | (5) | CO2 |
| | b. | Explain the Relationship types with example. | (5) | CO2 |
| 5. | a. | Consider the relational table given below and assess about the following SQL queries. Employee (Empno, Name, Department, Salary).
(i) List all the employees whose name starts with the letter 'L'.
(ii) Find the maximum salary given to employees in each department.
(iii) Find the number of employees working in 'accounts' department.
(iv) Find the second maximum salary from the table. | (5) | CO5 |
| | b. | Explain the join operations in relational algebra with an example | (5) | CO2 |

(OR)

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| 6. | a. | Write notes on Embedded SQL. | (5) | CO5 |
| | b. | Describe the aggregate functions in SQL with an example. | (5) | CO5 |
| 7. | a. | Briefly discuss about the functional dependency concepts. | (5) | CO4 |
| | b. | With suitable example explain the anomalies in Database. | (5) | CO4 |

(OR)

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| 8. | a. | Write the Armstrong rules? | (5) | CO4 |
| | b. | Explain 4NF with an example. | (5) | CO4 |
| 9. | a. | Brief about ACID Vs BASE Transactions. | (5) | CO3 |
| | b. | Write notes on Lock Based Protocols. | (5) | CO3 |

(OR)

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| 10. | a. | Explain View Serializability. | (5) | CO3 |
| | b. | Write notes on Shadow Paging. | (5) | CO3 |

Part C - Answer ALL Questions

1 x 15 = 15 Marks

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| 11. | a. | Assume we have the following application that models soccer teams, the games they play, and the players in each team. In the design, we want to capture the following:
<ul style="list-style-type: none"> • We have a set of teams, each team has an ID (unique identifier), name, main stadium, and to which city this team belongs. • Each team has many players, and each player belongs to one team. Each player has a number (unique identifier), name, DoB, start year, and shirt number that he uses. • Teams play matches, in each match there is a host team and a guest team. The match takes place in the stadium of the host team. • For each match we need to keep track of the following: <ul style="list-style-type: none"> o The date on which the game is played o The final result of the match o The players participated in the match. For each player, how many goals he scored, whether or not he took yellow card, and whether or not he took red card. o During the match, one player may substitute another player. We want to capture this substitution and the time at which it took place. • Each match has exactly three referees. For each referee we have an ID (unique identifier), name, DoB, years of experience. One referee is the main referee and the other two are assistant referee. Design an ER diagram to capture the above requirements. State any assumptions you have that affect your design. Make sure cardinalities and primary keys are clear. | (10) | CO2 |
| | b. | State concurrency control and explain the three problem statements. | (5) | CO3 |
| | | (OR) | | |
| 12. | a. | Explain the storage structure in DBMS. | (8) | CO3 |
| | b. | Identify the highest Normal Form in the Relation: $R(A,B,C,D,E)$
FD: $AE \rightarrow BC, AC \rightarrow D, CD \rightarrow BE, D \rightarrow E$ | (7) | CO4 |



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**FIFTH SEMESTER
ITA115 OBJECT ORIENTED ANALYSIS AND DESIGN**

Time: Three Hours

Max. Marks: 75

Part A - Answer ALL Questions

5 x 2 = 10 Marks

No.	Question	CO
i.	What is meant by Object Oriented Analysis and Design?	CO1
ii.	Define OCL.	CO2
iii.	Explain Design Pattern.	CO3
iv.	State the advantages of Test Driven Development.	CO4
v.	Name any 5 GRASP Principles.	CO5

Part B - Answer ALL Questions

5 x 10 = 50 Marks

No.	Question	Marks	CO
1.	a. Explain the Unified Process in detail.	(5)	CO1
	b. Discuss Class Modelling with suitable example. (OR)	(5)	CO1
2.	a. Draw a Class Diagram for Student Information System	(5)	CO1
	b. Draw a State Diagram for Bank Management System	(5)	CO1
3.	a. Discuss Object Constraint Language	(4)	CO2
	b. Explain System Sequence Diagram (OR)	(6)	CO2

- 10 or 2
3 or 11
5 or 6
1 or 8
2 or 10
10 or 12
- | | | | | |
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| 4. | a. | Explain Domain Model with example. | (5) | CO2 |
| | b. | Draw a Sequence Diagram for ATM System | (5) | CO2 |
| 5. | a. | Discuss Design Pattern. | (5) | CO3 |
| | b. | Explain Model View Control Pattern and its advantages. | (5) | CO3 |
| (OR) | | | | |
| 6. | a. | Explain GRASP pattern. | (5) | CO3 |
| | b. | Discuss any two GRASP Principles with example | (5) | CO3 |
| 7. | a. | Discuss Test Driven Development | (5) | CO4 |
| | b. | Explain Refactoring and its advantages | (5) | CO4 |
| (OR) | | | | |
| 8. | a. | Discuss UML Tool and UML as a blueprint. | (5) | CO4 |
| | b. | Explain the advantages of Test Driven Development | (5) | CO4 |
| 9. | a. | Discuss any two GRASP principles with example. | (5) | CO5 |
| | b. | Explain Package design | (5) | CO5 |
| (OR) | | | | |
| 10. | a. | Discuss Architectural Analysis and its advantages. | (5) | CO5 |
| | b. | Explain Logical Architecture | (5) | CO5 |

Part C - Answer ALL Questions

1 x 15 = 15 Marks

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|------|----|---|-----|-----|
| 11. | a. | Draw a Use Case Diagram for Bank Management System | (8) | CO1 |
| | b. | Draw a Activity Diagram for Exam Results of a Student | (7) | CO2 |
| (OR) | | | | |
| 12. | a. | Draw a Sequence Diagram for Inventory System | (7) | CO2 |
| | b. | Discuss Object design with GRASP | (8) | CO3 |
